Project: Worked examples
Title: Psychrometrics
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1. Scope and method

The state of moist air (humidity ratio, relative humidity, dew point, wet-bulb, enthalpy, specific volume, density) from dry-bulb and one other property at a stated pressure or altitude, and one air-handling process from that state with its heat and moisture balance: sensible heating or cooling, cooling with dehumidification, steam humidification, adiabatic humidification, or mixing with a second stream.

WHAT IT CALCULATES	RELATION
Saturation vapour pressure at the dry-bulb temperature, over water at or above 0 °C and over ice below	$p_{ws}(t)$ by Hyland–Wexler
Moisture content (humidity ratio) of the air	$W = 0.621945\ p_w/(p - p_w)$
Relative humidity	$\varphi = p_w/p_{ws}$
Specific enthalpy per kilogram of dry air	$h = 1.006\ t + W(2501 + 1.86\ t)$ kJ/kg$_{da}$
Specific volume	$v = 0.287042\ (t + 273.15)(1 + 1.607858\ W)/p$
Dew point and wet-bulb temperatures, by bisection (wet-bulb from the adiabatic-saturation relation)	$p_{ws}(t_{dp}) = p_w$
Sensible heating or cooling at constant moisture content	$Q = \dot{m}\ (h_2 - h_1)$
Cooling with dehumidification, less the heat leaving in the condensate	$Q_{total} = \dot{m}(h_1 - h_2) - \dot{m}_w\ h_w$, with $\dot{m}_w = \dot{m}(W_1 - W_2)$
Sensible and latent split of the coil load, along the path $t_1 \rightarrow t_2$ at W_1 then $W_1 \rightarrow W_2$ at t_2	$Q_{sensible} = \dot{m}(h_1 - h(t_2, W_1))$; $Q_{latent} = \dot{m}(h(t_2, W_1) - h_2)$; $SHR = Q_{sensible}/(Q_{sensible} + Q_{latent})$. With RH_2 blank and t_2 above the entering dew point the coil is dry: $W_2 = W_1$, no condensate, sensible only
Steam humidification	$h_2 = h_1 + (W_2 - W_1)\ h_{steam}$
Adiabatic (spray) humidification, at constant enthalpy	$h_2 = h_1$
Mixing two airstreams, weighted by dry-air mass flow	$W_{mix} = \Sigma\dot{m}\ W/\Sigma\dot{m}$; $h_{mix} = \Sigma\dot{m}\ h/\Sigma\dot{m}$

Basis: Psychrometrics (ideal-gas relations, processes)[1](Ch. 1); Saturation pressure of water and ice[2]; Properties of humid air (principles)[3](Section 1).

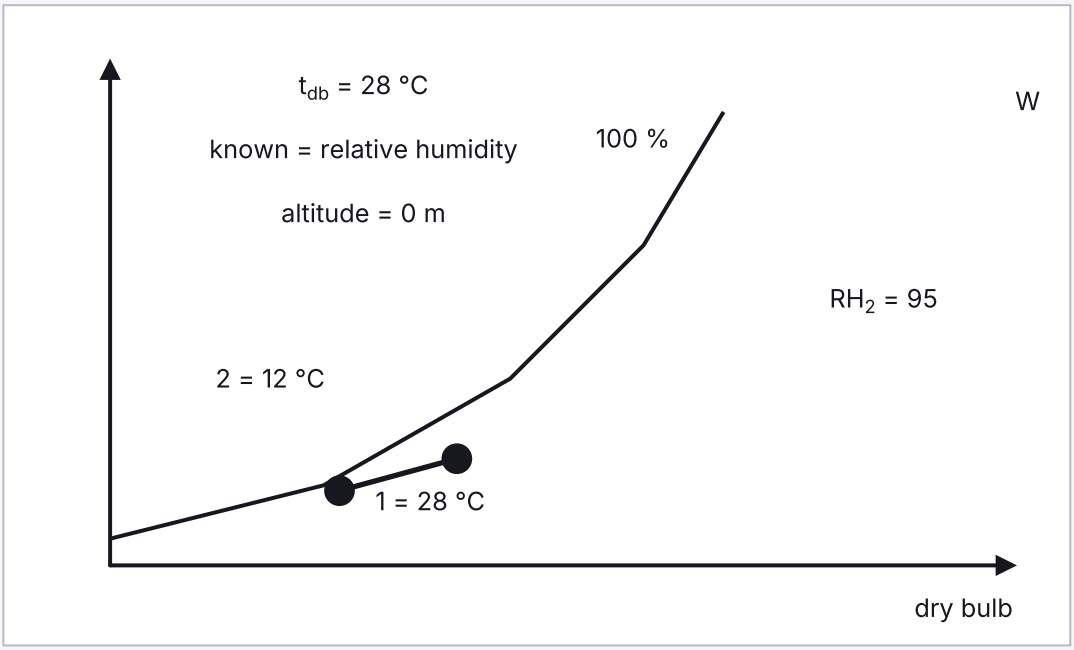


Figure 1 — Cooling with dehumidification: the state cools to t_2 at RH_2, losing moisture. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
altitude	Altitude (used when pressure is blank)	0	m	assumption: Sea-level site.	-500 … 5000 m
t_{db}	Dry-bulb temperature	28	°C	spec: CIBSE Guide A summer design condition for the site, assumed	-40 … 60 °C
known	Second known property	relative humidity		assumption: RH given.	any
RH	Relative humidity	60		spec: Design condition	0 … 100
m_{da}	Dry-air mass flow	2.4	kg/s	spec: AHU schedule AHU-01: 2.0 m³/s ≈ 2.4 kg/s dry air	0 … 1000 kg/s
process	Process	cooling with dehumidification		assumption: Chilled-water coil.	any
t_2	Leaving dry-bulb (sensible, cooling)	12	°C	spec: AHU schedule AHU-01 off-coil 12 °C	-40 … 80 °C
RH_2	Leaving relative humidity (cooling, evaporative)	95		assumption: Off-coil 95 % RH (typical contact factor).	0 … 100
h_{steam}	Steam enthalpy (steam humidification)	2676		default	2500 … 3000

3. Assumptions

A1. Moist air is an ideal mixture of ideal gases; the enhancement factor is taken as 1. (ASHRAE Fundamentals Ch. 1 ideal-gas relations.)
A2. Condensate leaves at the coil leaving dry-bulb; steam adds enthalpy h_{steam} at negligible mass of dry air. (Process conventions.)
A3. Adiabatic humidification follows a line of constant enthalpy (close to constant wet-bulb). (Evaporative process idealisation.)
A4. altitude (Altitude (used when pressure is blank)) is taken as 0 m. (Sea-level site.)
A5. known (Second known property) is taken as relative humidity. (RH given.)
A6. process (Process) is taken as cooling with dehumidification. (Chilled-water coil.)
A7. RH_2 (Leaving relative humidity (cooling, evaporative)) is taken as 95. (Off-coil 95 % RH (typical contact factor).)

4. Working

[2] p = 101.325 (1 - 2.25577e-5 z)^5.2559 [1](Ch. 1 eq. 3)
= z = 0 m
= 101.32 kPa
Barometric pressure (standard atmosphere)
[6] $p_{ws,1}$ = p_{ws}(t) Hyland–Wexler (entering)
= t = 28 °C
= 3.7822 kPa
Saturation pressure, state 1
[7] $p_{w,1}$ = p W/(0.621945 + W)
= W = 0.014249
= 2.2693 kPa
Vapour pressure, state 1
[8] $t_{1,out}$ = dry-bulb (entering)
=
= 28 degC
Dry-bulb, state 1
[9] W_1 = 0.621945 p_w/(p - p_w)
= 2.2693 / (101.32 - 2.2693)
= 14.249
Humidity ratio, state 1 (g/kg dry air)
[10] RH_1 = p_w / p_{ws}
= 2.2693 / 3.7822
= 60
Relative humidity, state 1 (%)
[11] φ_1 = p_w / p_{ws}
= 2.2693 / 3.7822
= 60 %
Relative humidity, state 1, as a percentage quantity (the same value as RH_1; a plain-number output cannot be linked to a percentage input)
[12] $t_{dp,1}$ = p_{ws}(t_{dp}) = p_w, bisection
= p_w = 2.2693 kPa
= 19.514 degC
Dew point, state 1
[13] $t_{wb,1}$ = adiabatic-saturation relation, bisection
= t = 28, W = 0.014249
= 22.074 degC
Wet-bulb, state 1
[14] h_1 = 1.006 t + W (2501 + 1.86 t)
= 1.006 × 28 + 0.014249 × (2501 + 1.86 × 28)
= 64.546

Enthalpy, state 1 (kJ/kg dry air)

[15] v_1 = 0.287042 (t + 273.15)(1 + 1.607858 W)/p

= p = 101.32 kPa

= 0.87267 m³/kg

Specific volume, state 1 (per kg dry air)

[16] ρ_1 = (1 + W)/v

=

= 1.1622 kg/m³

Moist-air density, state 1

[21] $p_{ws,2}$ = p_{ws}(t) Hyland–Wexler (leaving)

= t = 12 °C

= 1.4026 kPa

Saturation pressure, state 2

[22] $p_{w,2}$ = p W/(0.621945 + W)

= W = 0.0082878

= 1.3325 kPa

Vapour pressure, state 2

[23] $t_{2,out}$ = dry-bulb (leaving)

=

= 12 degC

Dry-bulb, state 2

[24] W_2 = 0.621945 p_w/(p − p_w)

= 1.3325 / (101.32 − 1.3325)

= 8.2878

Humidity ratio, state 2 (g/kg dry air)

[25] RH_2 = p_w / p_{ws}

= 1.3325 / 1.4026

= 95

Relative humidity, state 2 (%)

[26] φ_2 = p_w / p_{ws}

= 1.3325 / 1.4026

= 95 %

Relative humidity, state 2, as a percentage quantity (the same value as RH_2; a plain-number output cannot be linked to a percentage input)

[27] $t_{dp,2}$ = p_{ws}(t_{dp}) = p_w, bisection

= p_w = 1.3325 kPa

= 11.225 degC

Dew point, state 2

[28] $t_{wb,2}$ = adiabatic-saturation relation, bisection

= t = 12, W = 0.0082878

= 11.553 degC

Wet-bulb, state 2

[29] h_2 = 1.006 t + W (2501 + 1.86 t)

= 1.006 × 12 + 0.0082878 × (2501 + 1.86 × 12)

= 32.985

Enthalpy, state 2 (kJ/kg dry air)

[30] v_2 = 0.287042 (t + 273.15)(1 + 1.607858 W)/p

= p = 101.32 kPa

= 0.81856 m³/kg

Specific volume, state 2 (per kg dry air)

[31] ρ_2 = (1 + W)/v

=

= 1.2318 kg/m³

Moist-air density, state 2

[32] $m_w = \dot{m} (W_1 - W_2)$
$= 2.4 \times (14.249 - 8.2878)\ g/kg$
$= 0.014306\ kg/s$
Condensate
[33] $h_w = 4.186\ t_2$ (condensate enthalpy)
$= 4.186 \times 12$
$= 50.232$
Condensate enthalpy, kJ/kg
[34] $Q_{total} = \dot{m} (h_1 - h_2) - \dot{m}_w\ h_w$
$= 2.4 \times (64.546 - 32.985) - 0.014306 \times 50.232$
$= 75.027\ kW$
Total cooling load
[35] $Q_{sensible} = \dot{m} (h_1 - h(t_2, W_1))$
$= 2.4 \times (64.546 - 48.026)$
$= 39.648\ kW$
Sensible part: the air cooled from t_1 to t_2 at the entering humidity ratio W_1
[36] $Q_{latent} = \dot{m} (h(t_2, W_1) - h_2)$
$= 2.4 \times (48.026 - 32.985)$
$= 36.098\ kW$
Latent part: the moisture removed from W_1 to W_2 at t_2
[37] $SHR = Q_{sensible} / (Q_{sensible} + Q_{latent})$
=
$= 0.52344$
Sensible heat ratio

5. Results

SYMBOL	VALUE	DESCRIPTION
$t_{1,out}$	28 °C	Dry-bulb 1
$p_{ws,1}$	3.782 kPa	p_{ws} 1
$p_{w,1}$	2.269 kPa	p_w 1
W_1	14.249	W 1 (g/kg)
RH_1	60	RH 1 (%)
φ_1	60 %	φ 1 (RH as a % quantity, for linking)
$t_{dp,1}$	19.51 °C	Dew point 1
$t_{wb,1}$	22.07 °C	Wet-bulb 1
h_1	64.546	h 1 (kJ/kg)
v_1	0.8727 m³/kg	v 1
ρ_1	1.162 kg/m³	ρ 1
$t_{2,out}$	12 °C	Dry-bulb 2
$p_{ws,2}$	1.403 kPa	p_{ws} 2
$p_{w,2}$	1.332 kPa	p_w 2
W_2	8.2878	W 2 (g/kg)
RH_2	95	RH 2 (%)
φ_2	95 %	φ 2 (RH as a % quantity, for linking)
$t_{dp,2}$	11.22 °C	Dew point 2

SYMBOL	VALUE	DESCRIPTION
$t_{wb,2}$	11.55 °C	Wet-bulb 2
h_2	32.985	h 2 (kJ/kg)
v_2	0.8186 m³/kg	v 2
ρ_2	1.232 kg/m³	ρ 2
m_w	0.01431 kg/s	Condensate
Q_{total}	75.03 kW	Total cooling
$Q_{sensible}$	39.65 kW	Sensible cooling
Q_{latent}	36.10 kW	Latent cooling
SHR	0.52344	Sensible heat ratio

6. Checks (governing first)

None.

7. Applicability

None.

Defaults used (inputs left empty):
h_{steam}: not entered, so the default of 2676 is used.

8. Verification

```
Template bs.psychrometrics@1.3.0 Content hash 680e2f365a72ef0a Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. American Society of Heating, Refrigerating and Air-Conditioning Engineers. ASHRAE Handbook—Fundamentals. ASHRAE.
2. Hyland RW, Wexler A. Formulations for the thermodynamic properties of the saturated phases of H2O from 173.15 K to 473.15 K. ASHRAE Trans. 1983;89(2A):500-519.
3. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.